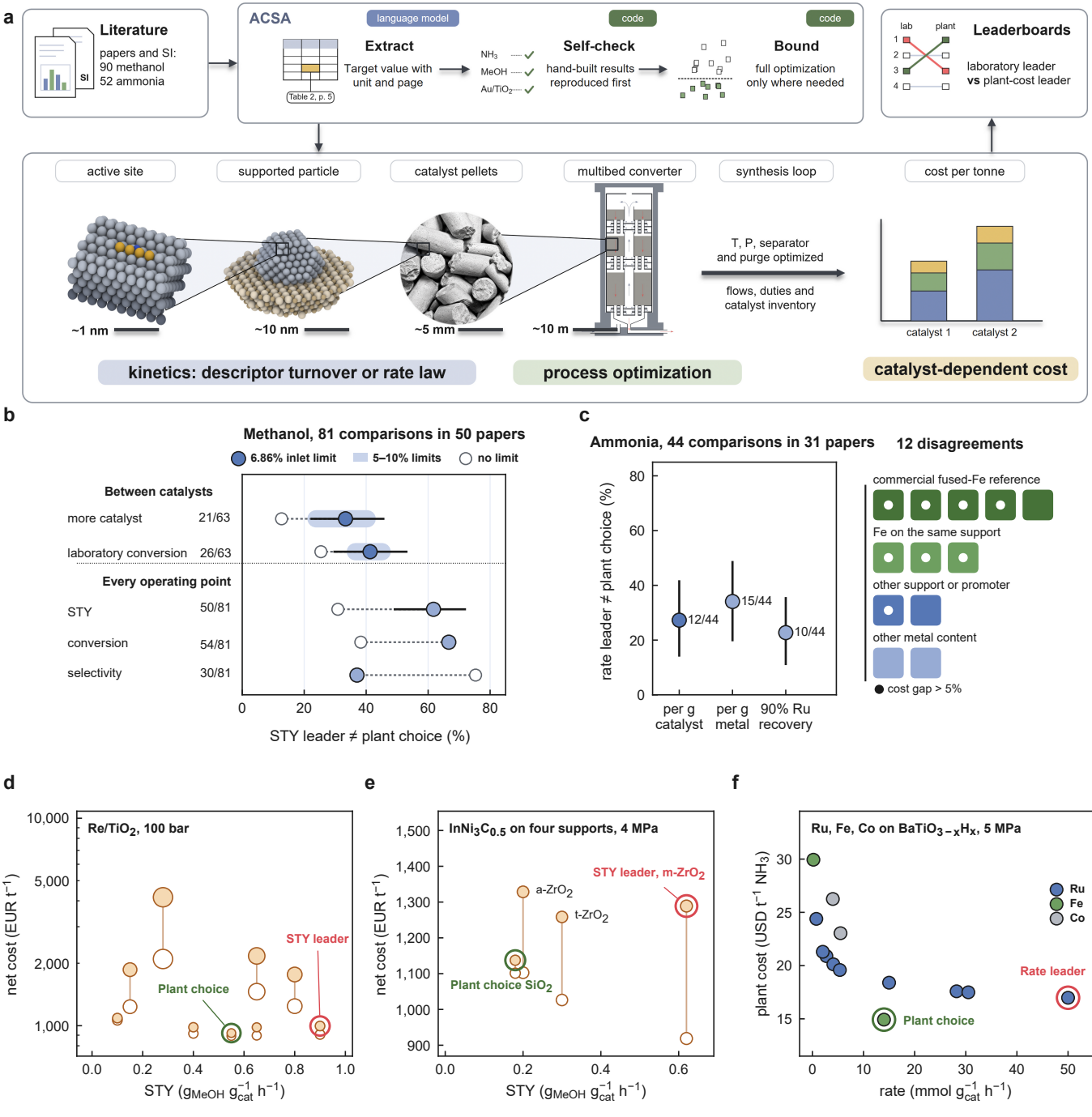

a
Literature
papers and SI:
90 methanol
52 ammonia
SI
ACSA
language model
code
code
Extract
Target value with unit and page
Table 2, p. 5
NH3
MeOH
Au/TiO2
Self-check
hand-built results reproduced first
Bound
full optimization only where needed
lab
plant
Leaderboards
laboratory leader
vs plant-cost leader
active site
supported particle
catalyst pellets
multibed converter
synthesis loop
cost per tonne
~1 nm
~10 nm
~5 mm
~10 m
T, P, separator and purge optimized
flows, duties and catalyst inventory
catalyst 1
catalyst 2
kinetics: descriptor turnover or rate law
process optimization
catalyst-dependent cost
b
Methanol, 81 comparisons in 50 papers
6.86% inlet limit
5–10% limits
no limit
Between catalysts
more catalyst 21/63
laboratory conversion 26/63
Every operating point
STY 50/81
conversion 54/81
selectivity 30/81
STY leader ≠ plant choice (%)
c
Ammonia, 44 comparisons in 31 papers
12 disagreements
rate leader ≠ plant choice (%)
12/44
15/44
10/44
per g catalyst
per g metal
90% Ru recovery
commercial fused-Fe reference
Fe on the same support
other support or promoter
other metal content
cost gap > 5%
d
Re/TiO2, 100 bar
net cost (EUR t−1)
STY leader
Plant choice
STY (gMeOH gcat−1 h−1)
e
InNi3C0.5 on four supports, 4 MPa
net cost (EUR t−1)
STY leader, m-ZrO2
a-ZrO2
t-ZrO2
Plant choice SiO2
STY (gMeOH gcat−1 h−1)
f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
plant cost (USD t−1 NH3)
Ru
Fe
Co
Rate leader
Plant choice
rate (mmol gcat−1 h−1)